Assembly-level recovery of the IVM-t1 mitochondrial-maintenance gene panel

tBLASTn comparison of T. brucei reference proteins against the assembled T. equiperdum IVM-t1 genome

Gene	Functional role	Best IVM-t1 hit	Alignment (aa)	Query coverage (%)	Identity (%)	E-value	Bit score	Assembly-level interpretation
ATPF1A	F_1-ATPase α subunit	CM010682.1	404	77.8	**25.99**	1.67×10^{-20}	96.3	Anomalous / unresolved
ATPF1B	F_1-ATPase β subunit	CM010674.1	519	100.0	**100.00**	0.00	969.0	Strong full-length recovery
ATPF1G	F_1-ATPase γ subunit	CM010681.1	305	100.0	**99.34**	0.00	625.0	Strong full-length recovery
ATOM40	Mitochondrial outer-membrane protein	CM010680.1	351	100.0	**100.00**	0.00	700.0	Strong full-length recovery
POLIB	Mitochondrial DNA polymerase	QSBY01000015.1	1,404	100.0	**99.93**	0.00	2,882.0	Strong full-length recovery
TAC102	Tripartite attachment complex protein	CM010678.1	648 + 219	91.2	**99.38**	0.00	1,300.0	Strong high-identity recovery; incomplete reference coverage

Note: Query coverage is calculated relative to the full-length reference protein.

ATPF1A lacked a convincing full-length, high-identity genomic match; its strongest alignment was only 25.99% identical over 404 aa. In contrast, ATPF1B, ATPF1G, ATOM40 and POLIB showed near-complete or complete high-identity recovery. TAC102 was represented by two high-identity alignments covering 867 of 951 aa (91.2% combined query coverage).

tBLASTn, translated nucleotide BLAST; IVM-t1, *Trypanosoma equiperdum* IVM-t1.